



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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ASSIGNMENT No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable realworld examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.



Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Input

```
1 package exp7;
2 class Shape
3 {
4     void area()
5     {
6         System.out.println("Area of Shape");
7     }
8 }
9 class Circle extends Shape
10 {
11     double radius = 5;
12     void area()
13     {
14         double area = 3.14 * radius * radius;
15         System.out.println("Area of Circle = " + area);
16     }
17 }
18 class Rectangle extends Shape
19 {
20     double length = 10;
21     double width = 5;
22     void area()
23     {
24         double area = length * width;
25         System.out.println("Area of Rectangle = " + area);
26     }
27 }
28 public class ShapeApplication
29 {
30     public static void main(String[] args)
31     {
32         Circle c = new Circle();
33         Rectangle r = new Rectangle();
34         c.area();
35         r.area();
36     }
37 }
```

terminated> ShapeApplication [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (01-Aug-2026, 1:36:05)
rea of Circle = 78.5
Area of Rectangle = 50.0

Output

```
Area of Circle = 78.5
Area of Rectangle = 50.0
```



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2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Input

```
ECommerceSystem.java X
1 package exp7;
2 @interface Product
3 {
4     void displayProduct();
5 }
6 @class CommonProduct
7 {
8     int id;
9     String name;
10    double price;
11    CommonProduct(int id, String name, double price)
12    {
13        this.id = id;
14        this.name = name;
15        this.price = price;
16    }
17 }
18 class Electronic extends CommonProduct implements Product
19 {
20    Electronic(int id, String name, double price)
21    {
22        super(id, name, price);
23    }
24    public void displayProduct()
25    {
26        System.out.println("Electronic Product");
27        System.out.println("ID: " + id);
28        System.out.println("Name: " + name);
29        System.out.println("Price: " + price);
30    }
31 }
32 class Clothing extends CommonProduct implements Product
33 {
34    Clothing(int id, String name, double price)
35    {
36        super(id, name, price);
37    }
38 }
```

Problems @ Javadoc Declaration Console X Properties
terminated> ECommerceSystem [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (01-Aug-2026, 1:45:13)
Electronic Product

Writable Smart Insert 73 : 2 : 1884

ENG IN 57% 13:45 01-08-2026

```
ECommerceSystem.java X
38    public void displayProduct()
39    {
40        System.out.println("Clothing Product");
41        System.out.println("ID: " + id);
42        System.out.println("Name: " + name);
43        System.out.println("Price: " + price);
44    }
45 }
46 class Grocery extends CommonProduct implements Product
47 {
48    Grocery(int id, String name, double price)
49    {
50        super(id, name, price);
51    }
52    public void displayProduct()
53    {
54        System.out.println("Grocery Product");
55        System.out.println("ID: " + id);
56        System.out.println("Name: " + name);
57        System.out.println("Price: " + price);
58    }
59 }
60 public class ECommerceSystem
61 {
62    public static void main(String[] args)
63    {
64        Electronic e = new Electronic(101, "Laptop", 55000);
65        Clothing c = new Clothing(102, "T-Shirt", 800);
66        Grocery g = new Grocery(103, "Rice", 1200);
67        e.displayProduct();
68        System.out.println();
69        c.displayProduct();
70        System.out.println();
71        g.displayProduct();
72    }
73 }
```

exp7
Product
CommonProduct
Electronic
Clothing
Grocery
ECommerceSystem
main(String[]) : void



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Output

Electronic Product

ID: 101

Name: Laptop

Price: 55000.0

Clothing Product

ID: 102

Name: T-Shirt

Price: 800.0

Grocery Product

ID: 103

Name: Rice

Price: 1200.0